

Radios coordinate today's swarms.

AirDnD removes them.

WHAT EXISTS TODAY

Central command or RF mesh

Shared tracks, assignment messages and handoffs coordinate each interceptor.

Broadband jamming breaks the coordinator first.



WHAT AIRDND ADDS

Coordination without shared truth

Each drone observes motion, infers friendly intent and decides from its own noisy local view.

Zero ground links · zero mesh · zero assignment messages

Hundreds of low-cost threats. Total RF + GNSS blackout. Intelligent interception without communication.

One task: defend a saturated corridor

PRIMARY USER

RSAF Air Defence and Operations Command

Air Surveillance and Control Group and coastal air-defence batteries.

TODAY

Allocate every engagement

Operators and RF C2 distribute tracks, assign weapons and coordinate a restrike.



WITH AIRDND

Define the corridor; supervise outcomes

Interceptors distribute targets, recover from misses and preserve separation locally.

Learned beliefs. Inspectable decisions.



$$U_{ij} = \text{consequence} \cdot P(\text{leak}) \cdot P(\text{success}) \cdot [1 - P(\text{covered})] - \text{costs}$$

PRIVATE COVERAGE WINDOWS

DETERMINISTIC TIE-BREAKING

SWITCHING HYSTERESIS

REVERSE-INS RTH

Measured against independent greedy pursuit

100

hostiles per run

125

interceptors

30

paired seeds

INDEPENDENT GREEDY

53.9%

mean leakage

-28.8 pp

53.4% relative reduction

AIRDND

25.1%

mean leakage

0 simulated friendly collisions

59.58 m mean minimum separation

19.2 ms p95 full simulation

Simulation evidence • fixed configuration • exact seeds, raw logs and SHA-256 manifest archived

What works—and what does not yet

BIGGEST DOUBT

Optical track stability in heavy tropical rain

AirDnD depends on seeing hostile and friendly motion. Dense rain, glare, smoke or aerosol obscurants can degrade intent inference into dead reckoning.

Simulation only	No physical flight or aerodynamic validation.
Sensor models are stubbed	NIR loss and MEMS drift are sampled, not bench-measured.
Not zero leakage	25.1% mean leakage remains at 100-hostile scale.
Saturation breaks reserves	Above the modeled density, refill capacity is exhausted.

Claimed: software mechanism + simulation evidence. Not claimed: combat readiness or flight-proven performance.

A software-defined autonomy payload

WHO PAYS

DSTA • DSO • MINDEF

Licensed through a sovereign integrator such as ST Engineering for low-cost interceptor fleets.

≈ S\$1,150

modeled airframe + compute + dual optical seekers



We will keep building it: from coordinated software proof to measured hardware autonomy under blackout.